

SHL Assessment Recommender

OVERVIEW & PURPOSE

The SHL Assessment Recommender is a stateless conversational agent that helps hiring managers select SHL assessments through multi-turn dialogue. Powered by a 7-node LangGraph agent and exposed via FastAPI, it uses hybrid retrieval to match hiring intents and accumulated constraints (e.g., role, skills, duration) to a normalized catalog of 377 products.

OBJECTIVES

- 1. **Accurate Assessment Matching:** Map open-ended queries to relevant assessments using hybrid semantic and keyword retrieval.
- 2. **Multi-Turn Constraint Accumulation:** Persist and refine user constraints across conversation turns without requiring repetition.
- 3. **Grounded, Hallucination-Free Recommendations:** Strictly source all suggestions from the retrieved catalog to eliminate hallucinations.
- 4. **Schema-Safe Output:** Enforce Pydantic validation, multi-tier slug correction, and safety checks on every response.

ARCHITECTURE & CORE PIPELINE

The system has two main subsystems: **Ingestion Pipeline** and a **Conversational Agent**.

A. Ingestion Pipeline

Transforms raw scraped SHL catalog data into three retrieval indices:

Output	Purpose
catalog.json	377 deduplicated, normalized assessments keyed by slug. Duration normalized to integer minutes; missing values set to null.
bm25_indices.pkl	Two BM25Okapi indices — one on enriched text chunks (name repeated 3× + keys + description), one on names only. Name repetition boosts exact-name match scores.
Pinecone shl-assessments	Dense 384-dim cosine vector index (all-MiniLM-L6-v2) with metadata filters for duration and remote support. Slugs are the universal ID across all three stores.

B. Conversational Agent

A 7-node LangGraph state machine exposed via FastAPI at POST /chat. Nodes execute conditionally based on intent and turn count:

Node	Purpose
A : Turn Budget	Counts turns; forces recommendation at turn 6+
B : Gatekeeper	Intent classifier: VAGUE / CONSTRAINED / REFINE / COMPARE / OUT_OF_SCOPE
C : Role Classifier	Generates 3–5 search queries across knowledge, personality, cognitive, SJT
D : State Extractor	Accumulates constraints (role, seniority, skills, language, duration) across turns
E : Retriever	Hybrid multi-query retrieval with RRF merge + keyword boost → top-15 to synthesizer
F : Synthesizer	Generates 3–10 grounded JSON recommendations; falls back if Groq rate-limits
G : Schema Validator	Pydantic validation, slug correction, fuzzy name matching, safety checks

C. Retrieval Strategy

Implements a hybrid retrieval pipeline to surface a final top-15 candidate list.

- **Methodology:** Merges Pinecone dense and BM25 sparse vectors utilizing Reciprocal Rank Fusion (RRF, k=60).
- **Query Facets:** Executes searches across five distinct vectors: enriched role classifications, raw user inputs, isolated skill parameters, foundational categories (OPQ personality and Verify G+ cognitive), and direct keyword boosts.

D. Synthesizer Prompt Design

- **Domain Integration:** Directly encodes SHL category definitions and mandatory battery-building rules into the system prompt.
- **Decision Logic:** Dynamically shifts behavior based on intent—triggering clarifying dialogue for vague queries, or committing 3–10 precise recommendations for highly constrained inputs.

E. Validator Safety Checks

Pre-processes outputs (list-to-string coercion) for strict Pydantic parsing. Ensures accuracy via **multi-tier slug matching** (**exact** → **case-insensitive** → **name-exact** → **substring**) and prevents conversation termination if zero recommendations are generated.

MODELS USED

Model	Purpose
all-MiniLM-L6-v2	Embedding generation for Pinecone dense index and query vectors (384-dim, cosine similarity)
gemini-flash-3-preview	Classification and extraction: Gatekeeper, Role Classifier, State Extractor nodes (temperature=0)
openai/gpt-oss-20b (Groq)	Primary synthesizer: structured JSON recommendation generation (max_tokens=4096, json_object mode)
gemini-2.5-pro	Fallback synthesizer: activates automatically if Groq rate-limits or errors

STANDOUT FEATURES FOR PRODUCTION

Features	How It Solves Production / Latency Constraints
Hybrid Retrieval + RRF	BM25 handles exact assessment names; dense vectors handle semantic intent. RRF fusion with k=60 balances both without needing manual score calibration.
5-Facet Multi-Query Fan-Out	A single user message spawns up to 5 distinct search queries, dramatically increasing the chance of surfacing the correct niche assessment.
Stateless Turn-Capped Agent +	Zero server-side session state; all conversation history is injected per-request. The turn budget hard-caps at 8 turns, ensuring the conversation never stalls.
LLM Fallback Chain	Groq (primary) → Gemini 2.5 Pro (fallback). If Groq rate-limits during peak load, the backend silently re-routes with no user-visible disruption.

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Github: https://github.com/hemangjain17/Hemang_shl_assessment

Deployed Link: <https://hemang-shl-assessment-4nye.vercel.app>

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